

HealthPulse AI - Architecture Documentation

Autonomous Multi-Hospital Patient Intake, Scheduling & Voice Agent Platform

1. System Architectural Overview

HealthPulse AI is engineered around a decoupled 5-tier architecture ensuring strict separation of concerns, comprehensive role-based multi-tenancy, and auditable healthcare workflow execution. The AI agent operates strictly as an administrative assistant with zero direct access to SQL databases or external Electronic Health Record (EHR) systems.

Tier / Layer	Components & Responsibilities
Client & Interface Layer	React SPA with Web Speech API for real-time STT and voice synthesis (TTS). Role-based dashboards for Patients, Doctors, Hospital Admins, and Platform Admins.
AI & Orchestration Layer	LangGraph state machine graph implementing safety guardrails, intent classification, structured context resolution, and controlled capability routing.
Core Services Layer	Deterministic scheduling engine (source of truth), appointment lifecycle manager with state transitions, pre-visit workflow executor, and structured audit logger.
Integration & Connector Layer	Standardized HealthcareSystemConnector interface communicating with external Mock EHR. Includes failure simulator (Timeout, Network Error, Slot Conflict, Auth Error, Fail).
Data & Persistence Layer	SQLAlchemy ORM with tenant-scoped models (Hospital, Department, Specialty, Doctor, Calendar, Availability, Appointment, State History, Questionnaire, Audit Log).

2. Multi-Tenant Isolation & Role-Based Access

Tenant isolation is enforced natively at the API gateway and SQLAlchemy query layer through security dependencies:

- **Platform Admin:** Global visibility across all hospitals, doctor calendars, and system-wide EHR operations.
- **Hospital Admin:** Strictly scoped to user hospital_id. Cannot view, create, or alter doctors, calendars, or operations of other hospital tenants.
- **Doctor:** Scoped to doctor_id and hospital_id. Can only review scheduled patients and pre-visit clinical questionnaire submissions.
- **Patient:** Scoped to patient_id. Can only view/manage personal appointments and submissions.

3. Appointment Finite State Machine & Audit History

All appointment state transitions must pass finite state validation via can_transition_appointment(). Every transition immutably logs into AppointmentStateHistory with actor email and correlation ID.

From Status	Allowed Target Statuses	Enforced Transition Logic
PENDING	CONFIRMED, FAILED, CANCELLED	Transition to CONFIRMED only upon external EHR verification.
CONFIRMED	RESCHEDULED, CANCELLED, COMPLETED	Active booking; can be rescheduled or cancelled with slot release.
RESCHEDULED	CONFIRMED, CANCELLED	Transitions back to CONFIRMED once new slot verified.
CANCELLED	Terminal State	Cannot be resurrected; slot made available to others.
COMPLETED	Terminal State	Post-visit final state.
FAILED	PENDING, CANCELLED	Encountered failure; allows idempotency-safe retry or cancellation.

4. Unknown Outcome & Network Timeout Recovery Flow

When an external EHR call encounters a network timeout (HTTP 504), HealthPulse AI never issues a blind retry (preventing double bookings). Instead, the engine transitions the booking to SYNCHRONIZATION_PENDING and initiates an out-of-band verification query by idempotency key. Upon locating the external booking reference, state is cleanly synchronized to CONFIRMED with zero duplicate creations.